

MATH 2211, SPRING 2026: MIDTERM 1

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name _____

	Score
Problem 1	/10
Problem 2	/10
Problem 3	/10
Problem 4	/10
Problem 5	/10
Total	/50

Name _____

- (1) (10 points) **True or false. Write one or two lines justifying your answer:**

(a) If F is a field and $a \in F$, then $(-1) \cdot a = -a$.

True. $(1 + (-1)) \cdot a = 1 \cdot a + (-1) \cdot a$
 $\parallel \parallel \Rightarrow (-1) \cdot a = -a$
 $0 \cdot a = 0$

(b) $\mathbb{Z}/6\mathbb{Z}$ with modular addition and multiplication is a field.

False $2 \cdot 3 = 0 \in \mathbb{Z}/6\mathbb{Z}$
 $\Rightarrow 2 \text{ \& } 3 \text{ are not invertible}$
 $\Rightarrow \mathbb{Z}/6\mathbb{Z} \text{ is not a field}$

(c) There exists a set of 5 linearly independent vectors inside $\mathbb{R}^4$.

False $\mathbb{R}^4$ admits a basis with 4 elements
 $\& \text{ so by the exchange lemma every}$
 $\text{list of lin. ind. vectors has } \leq 4 \text{ elements}$

(d) The equation $z^n = 3i + 4$ admits n distinct solutions for z in $\mathbb{C}$.

True The solutions are $5^{1/n} e^{i(\theta + 2\pi k)/n}$
 $0 \leq k \leq n-1$
 $\left[\begin{array}{l} 3i + 4 = 5e^{i\theta} \\ \text{where} \\ \cos(\theta) = \frac{3}{5}, \sin(\theta) = \frac{4}{5} \end{array} \right]$

(e) The polynomials $1+x, 1-x \in \text{Poly}(\mathbb{Z}/3\mathbb{Z})$ are linearly independent.

True For two vectors linear dependence
 would mean
 $1+x = c(1-x) \quad \exists c \in \mathbb{Z}/3\mathbb{Z}$
 $\Downarrow$
 $1+x = c - cx \Rightarrow \underbrace{c=1 \text{ \& } c=-1}_{\text{not possible!!}}$

Name _____

(2) (10 points) Show that, for any integer $n \geq 1$, we have

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$$

Here, $n!$ is n -factorial, defined recursively by $1! = 1$ and $n! = n \cdot (n-1)!$, so that, for $n \geq 2$, we have $n! = n(n-1) \cdots 2 \cdot 1$.

Proof by induction $P(n)$: equality holds for n Base step ($n=1$) $P(1)$

$$\begin{array}{ccc} 1 \cdot 1! & \stackrel{?}{=} & (1+1)! - 1 \\ || & & || \\ 1 & \stackrel{\checkmark}{=} & 2 - 1 \end{array}$$

Inductive hypothesis: $P(n)$

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$$

Inductive step: Add $(n+1) \cdot (n+1)!$ to both sides:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! + (n+1) \cdot (n+1)! = (n+1)! - 1 + (n+1)(n+1)!$$

$$\rightarrow = (n+1)! [1 + n+1] - 1$$

$$= (n+1)! \cdot (n+2) - 1$$

$$= (n+2)! - 1$$

 $\Rightarrow P(n+1)$ holds $\checkmark$

Name _____

(3) (10 points) Consider the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \mathbb{Q}^3$$

(a) Find all solutions for $x_1, x_2, x_3, x_4 \in \mathbb{Q}$ to the equation

$$x_1 v_1 + x_2 v_2 + x_3 v_3 + x_4 v_4 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix}$$

This is the system

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 1 & 1 & -1 & 1 & -2 \\ -1 & 1 & 3 & 0 & 0 \end{array} \right] \xrightarrow[R_3+R_1]{R_2-R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 0 & -1 & -1 & 1 & -5 \\ 0 & 3 & 3 & 0 & 3 \end{array} \right]$$

Row reduced system

$$x_1 - 2x_3 = 1$$

$$x_2 + x_3 = 1$$

$$x_4 = -4$$

$$\xrightarrow{R_3+2R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 0 & -1 & -1 & 1 & -5 \\ 0 & 0 & 3 & -2 & -12 \end{array} \right] \xrightarrow[(-1) \cdot R_2]{\frac{1}{3} \cdot R_3} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & -1 & 5 \\ 0 & 0 & 1 & -2 & -4 \end{array} \right] \xrightarrow[R_1-2R_2]{R_2+R_3} \left[\begin{array}{cccc|c} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 1 & -1 & 5 \\ 0 & 0 & 1 & -2 & -4 \end{array} \right]$$

$$\text{Solutions: } \left\{ (x_1, x_2, x_3, x_4) = (1+2t, 1-t, t, -4) \right\}_{t \in \mathbb{Q}}$$

(b) Can you find a subset of $\{v_1, v_2, v_3, v_4\}$ that is a basis for $\mathbb{Q}^3$?

Look at columns with leading 1s:

$$\{v_1, v_2, v_4\}$$

is linearly independent & hence
a basis for $\mathbb{Q}^3$.

Name _____

(4) (10 points) Consider the subset

$$U = \{f(x) \in \text{Poly}_{\leq 3}(\mathbb{R}) : \int_0^1 f(x) dx = 0\}.$$

(a) Show that U is an $\mathbb{R}$ -subspace of $\text{Poly}_{\leq 3}(\mathbb{R})$.

[See problem 2 on HW 4](#)

(b) Find a basis for U and compute $\dim(U)$.

Name _____

- (5) (10 points) Give a rigorous proof of the following assertion: If V is a finite dimensional F -vector space and $W \subset V$ is an F -subspace with $\dim W = \dim V$, then in fact $W = V$.

Choose bases $\{w_1, \dots, w_n\}$ for W ← linearly independent
 $\{v_1, \dots, v_n\}$ for V ← spans V

By the Steinitz Exchange Lemma, we can replace each v_i with a w_i (after reindexing) without changing the fact that the set spans V . Since $n=n$, this means that

$\{w_1, \dots, w_n\}$ spans V !

But by hypothesis their span is W .

$\Rightarrow$

$$W = V$$